

KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

The v6 reduction — wide calibration set, freely fitted, Gaia applied afterwards

Prepared 2026-09-09. Solution: `~/KMTdata/Results/v6/IFfinal_<field>.mat` (variable IFsys), with the tie in `Tie_<field>.mat`.

1. A short description

The analysis is restricted to a calibration set defined by **magnitude and isolation alone**: $14 < I < 19$ with no companion brighter than $I = 18$ within 2 arcsec. Every one of those stars is freely fitted and every one takes part in determining the per-epoch frame at every step. **No Gaia quality cut is applied at selection**; RUWE enters only afterwards, when the finished solution is tied to Gaia. The target passes the selection on its own merits and needs no special handling.

2. Object selection, step by step

step	BLG41	BLG01
raw sources in the MSc file	1177	1187
survive the standard quality cuts	594	621
have an OGLE I magnitude	540	478
in $14 < I < 19$	529	470
... and isolated: no $I < 18$ companion within 2.0 arcsec	287	253
= analysis set actually fitted	287	253

The isolation cut is the dominant one, removing 46% of the magnitude-selected sources on both fields.

The target is a member. At $I = 18.13$ it lies inside the window and passes the isolation test, so it enters the fit on its own merits rather than being added by hand.

For the tie only, Gaia is required: **273 of 287** and **225 of 253** sources have a Gaia counterpart, and of those **220** and **177** pass $\text{RUWE} < 1.4$. The tie is fitted on a narrower set still — those RUWE-clean stars that also fall in $15 < I < 17$, numbering **118** and **90** — and is then applied to every source. Section 3 gives the reason and what it is worth.

3. Gaia, applied only at the end

The tie set. The tie is fitted on stars that pass **$\text{RUWE} < 1.4$ and $15 < I < 17$** — 118 on BLG41 and 90 on BLG01 — chosen to be both astrometrically well behaved in Gaia and well measured here. It is then applied to every source in the field.

The map. Gaia standard coordinates are fitted directly onto cut-out pixels from those stars — not carried through OGLE, which would inherit its ~ 110 mas astrometric error. Median residual **23.4 mas (BLG41)** and **22.1 (BLG01)**, with

scales of 2.5195 and 2.5290 pix/arcsec against the 2.500 expected at 0.4"/pix. Restricting the tie set to $15 < l < 17$ improves the map by 35 to 40% over using all RUWE-clean stars, since bright stars have better positions.

The gauge, and a correction to all earlier work. A free fit determines astrometry only up to a gauge: applying a global affine to all positions and compensating in the per-epoch transformations changes no residual. In the proper motions this freedom is **a constant plus a linear gradient across the field** — six parameters, not two.

Removing only the constant leaves the gradient in place, and it shows up plainly in the cross terms: $\text{corr}(\Delta\mu_\alpha, Y) = -0.75$ on BLG41, a **shear of -0.037 mas/yr per pixel**, and -0.035 on BLG01 — nearly the same value in two independently reduced fields, so it is a property of the method, not noise. The full six-parameter gauge is therefore removed here:

scatter of our proper motions about Gaia	constant only	full gauge
BLG41	4.34 / 3.57 mas/yr	0.98 / 0.66
BLG01	3.48 / 3.20 mas/yr	1.00 / 0.70

A factor of three to four.

What the magnitude cut is worth. Evaluated on the same test set — all RUWE-clean stars, so neither tie is flattered by its own training sample — the narrower tie set changes the proper-motion agreement hardly at all:

scatter about Gaia, all RUWE-clean stars	tie on all RUWE-clean	tie on $15 < l < 17$
BLG41	0.979 / 0.664	1.022 / 0.640
BLG01	1.003 / 0.697	1.055 / 0.669

mas/yr — a few per cent, inconsistent in sign. The reason is that the tie was never the limiting term. The narrower set is better measured, 0.57 against 0.98 mas/yr per star, and it does determine the gauge slightly more precisely, **0.053 against 0.066 mas/yr**; but halving the number of stars nearly cancels that, and either way the gauge contributes **under 1% of the total variance**. The remaining 99% is each star's own proper-motion error, which no choice of tie stars can reach. The cut is kept because it clearly improves the positional map and costs nothing, not because it improves the motions.

4. The chain, and every effect fitted in it

Two steps, both on the 287 / 253 source set:

1. **Full-decade SysRem correction** — 2 components on the astrometric residuals, computed once and saved. 2. **Final solution**, reusing that correction.

	BLG41	BLG01
step 1	19.8 min	16.7 min
step 2	35.6 min	31.3 min
convergence over stars brighter than 17, last 3 iterations	0.0028 mas	0.0007 mas

Each step runs two passes internally. `IFsys.AnnualEffect` and `IFsys.PixPhase` read 0 in the saved solutions, which does **not** mean those effects were ignored: they are fitted and subtracted from the data in pass 1 and forced off in pass 2, which works on data from which they have already been removed.

	pass 1	between	pass 2 (final)
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per-epoch affine	fitted	fitted
per-source position and motion	fitted	fitted
DCR / chromatic refraction	fitted	fitted again
annual term	fitted and subtracted from the data	forced off
pixel-phase term	fitted and subtracted from the data	forced off
SysRem		applied
iterative reweighting	15 iterations	6 iterations

- **Per-epoch affine** — 6 parameters per exposure in `ParE`, stored as the deviation from the identity: translation (2), rotation (1), uniform scale (1), shear (2). Each epoch's six parameters come from 287 sources, so 574 equations for 6 unknowns.
- **Per source** — position at `JD0 = 2019-06-01` and proper motion, 4 parameters in `ParS`. All free; `ParSFixed` is empty.
- **DCR** — `[1, sin(pa)·secz, cos(pa)·secz]` plus six higher orders in `(secz·sin pa)n, (secz·cos pa)n` for `n = 2, 3, 4`: 9 per axis, 18 in total, fitted in 6 equal-population colour bins. Only the *differential* part is recoverable — refraction common to the field is exactly a translation and is absorbed by `t_x, t_y` every epoch.
- **Annual term** — a quartic in the phase of the calendar year, 10 parameters, fitted globally, one curve shared by every star. Not parallax, which is per-source and switched off.
- **Pixel phase** — a quintic in the sub-pixel phase, 10 parameters, global.
- **SysRem** — 2 components over the whole decade, reused rather than refitted per season, with point-wise rejection.
- **Weighting** — each source weighted by its own residual scatter, recomputed each iteration, with moving-median outlier rejection over 30 epochs.

No per-season fitting enters the solution. Seasons are used only for reporting.

5. Results

	BLG41 ΔX / ΔY	BLG01 ΔX / ΔY
stars brighter than <code>l = 17</code>	5.15 / 5.06	4.93 / 5.12
the target	23.75 / 19.63	21.36 / 20.37

mas. The calibration stars are measured to about 5 mas over the decade, and the target, at `l = 18.13`, to about 20 mas — consistent with the running median of the residual RMS at its magnitude.

6. Proper motion of the target, with uncertainties

	$\mu_\alpha \cos \delta$	μ_δ
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BLG41	-1.937 ± 0.335	-7.140 ± 0.360
BLG01	-1.658 ± 0.256	-7.089 ± 0.377
difference between fields	-0.279 ± 0.421	-0.052 ± 0.521

mas/yr. The two fields agree **within their errors**, at 0.66 and 0.10 sigma.

The quoted error is the **internal** one: the slope uncertainty from the scatter of the ten season means about the fitted line. It is deliberately not computed from the per-epoch count, which would give an absurdly small number — the nightly noise is strongly correlated. Median internal uncertainty over all sources is 0.127 / 0.114 (BLG41) and 0.123 / 0.112 (BLG01) mas/yr.

That internal error is still optimistic by a factor of two to three. The external check — the scatter of our proper motions about Gaia's, 1.02 / 0.64 and 1.06 / 0.67 mas/yr per star — is larger than the internal estimate would predict even after Gaia's own 0.15 mas/yr is allowed for. Use **~1.0 / 0.7 mas/yr** as the realistic per-star uncertainty; the internal figure captures the random part only and misses systematics that bias a star's slope while leaving its seasons internally consistent.

Gaia's own value for the target is $-2.479 / -9.129$ mas/yr. Our $\mu_\alpha \cos \delta$ agrees; our μ_δ differs by about 2 mas/yr, which is 3 sigma on the external error. The target is blended with a neighbour 4.5 pix away and its motion here is fitted straight through a multi-year magnification event, so this is not a clean comparison.

7. Comparison stars

Six, chosen to be the same physical objects in both fields — three near the target's own magnitude and three brighter:

tag	I	dist [pix]	decade RMS $\Delta X / \Delta Y$, BLG41	BLG01
m18_d04	18.14	4.5	29.03 / 27.91	24.84 / 26.86
m18_d17	18.09	17.3	51.68 / 27.85	85.47 / 28.47
m18_d48	18.05	48.4	17.92 / 16.76	14.58 / 15.57
m16_d24	16.68	24.4	4.19 / 4.27	4.17 / 4.38
m16_d30	16.63	30.4	6.91 / 5.85	6.43 / 5.55
m16_d31	16.75	30.6	8.48 / 7.79	11.20 / 7.71

mas. The bright trio reproduces between the fields almost exactly — m16_d24 gives $-3.31 / -7.81$ on BLG41 and $-3.31 / -7.89$ on BLG01 — while the $I \approx 18$ trio does not: m18_d17 gives -6.03 against -10.05 mas/yr, and its decade RMS reaches 52 and 85 mas with only ~2400 usable epochs. It is a poor star and should not be used as a control.

m18_d04 is the neighbour blended with the target, 4.5 pix away under ~7 pix seeing. It is a legitimate member of the calibration set — the target at $I = 18.13$ is fainter than the $I < 18$ companion threshold, so it does not trigger rejection — but its centroid tracks the target's brightening through the event, and it is not an independent object.

8. Chi-squared of the monthly binned residuals

For every star the residuals are binned in sidereal months; each bin contributes $(\text{mean/standard error})^2$ in both axes, and $\text{DoF} = 2 \cdot N_{\text{bins}} - 4$. Values are not divided by DoF.

	BLG41	BLG01
stars measured	287	253
median DoF	180	180
target χ^2	1574 (DoF 180)	1313 (DoF 180)
target χ^2/DoF	8.7	7.3
field median χ^2/DoF	7.2	7.1

The target is unremarkable. Its χ^2 sits in the body of the distribution, not in the tail: 8.7 against a field median of 7.2 on BLG41, and 7.3 against 7.1 on BLG01. Whatever astrometric excursion the event produces is not large enough to push the target out of the population of ordinary stars.

The field median of $\chi^2/\text{DoF} \approx 7$ is itself informative: it says the standard error of the mean within a month **underestimates the true bin uncertainty by about $\sqrt{7} \approx 2.6$** , because the noise within a month is correlated rather than independent. This is the same factor that makes the internal proper-motion errors of section 6 optimistic.

9. Overlap of the two fields

Stars are identified through the OGLE catalogue, which resolves about six times more sources than KMT detects; matching the two source lists directly is ambiguous because a close pair may be split in one field and merged in the other.

	BLG41	BLG01
sources	594	621
matched to an OGLE entry	551 (93%)	497 (80%)
distinct OGLE stars behind them	454	394

OGLE I	BLG41	BLG01	in both	distinct	found in both
< 15	28	22	22	28	78.6%
15 – 16	54	42	42	54	77.8%
16 – 17	172	149	146	175	83.4%
17 – 18	135	119	117	137	85.4%
18 – 19	55	44	41	58	70.7%
all brighter than 19	444	376	368	452	81.4%

About four stars in five are detected in both fields, flat from the brightest down to $I = 18$. **There is nothing fainter than $I = 19$ to compare:** our source lists span 12.87 to 18.98 in both fields, while OGLE itself reaches 21.40 and holds 2660 entries fainter than 19. That is the KMT detection limit in this crowding, not a catalogue cut.

10. Files

file	content
report_v6_RMS.png	residual RMS against OGLE I, per axis and field
report_v6_pm_vs_gaia.png	our absolute proper motion against Gaia's, after the tie
report_v6_chi2.png	χ^2 distribution of the monthly binned residuals, target marked
report_v6_radec_<field>_<tag>.png	RA and Dec residuals against time, and against each other colour-coded by time, removed
report_v6_motion_<field>_<tag>.png	the 2x2 motion figures, proper motion retained and removed
source_motion_v6_<field>.csv	the target, per epoch
source_motion_v6_<field>_<tag>.csv	each comparison star, per epoch

CSV columns are JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season, with positions relative to that star's own mean and **proper motion not removed**. Headers carry the star's magnitude, pixel position, sky position and absolute proper motion.

In the motion figures the columns are the two axes, **X left and Y right**; the top row keeps the proper motion and the bottom row removes it. The legend quotes the motion along the pixel axis and the heading quotes it on the sky. **Pixel X runs opposite to RA**, so those two carry opposite signs in X.

Figures

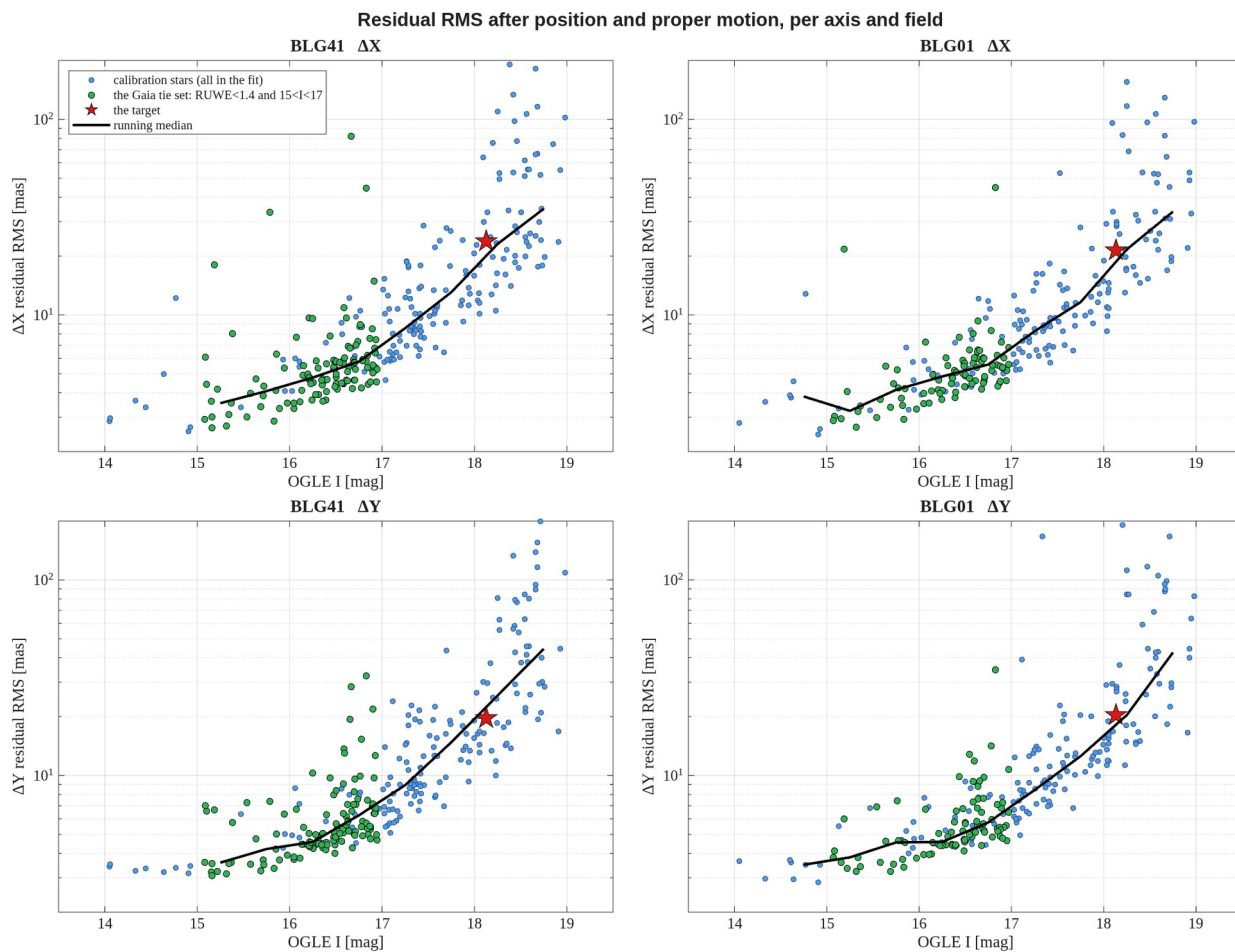


Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, after position and proper motion have been removed. Blue: all 287 / 253 calibration stars, every one freely fitted. Green: the subset with a Gaia RUWE below 1.4, used only for the tie. Red: the target. Black: running median.

report_v6_RMS.png

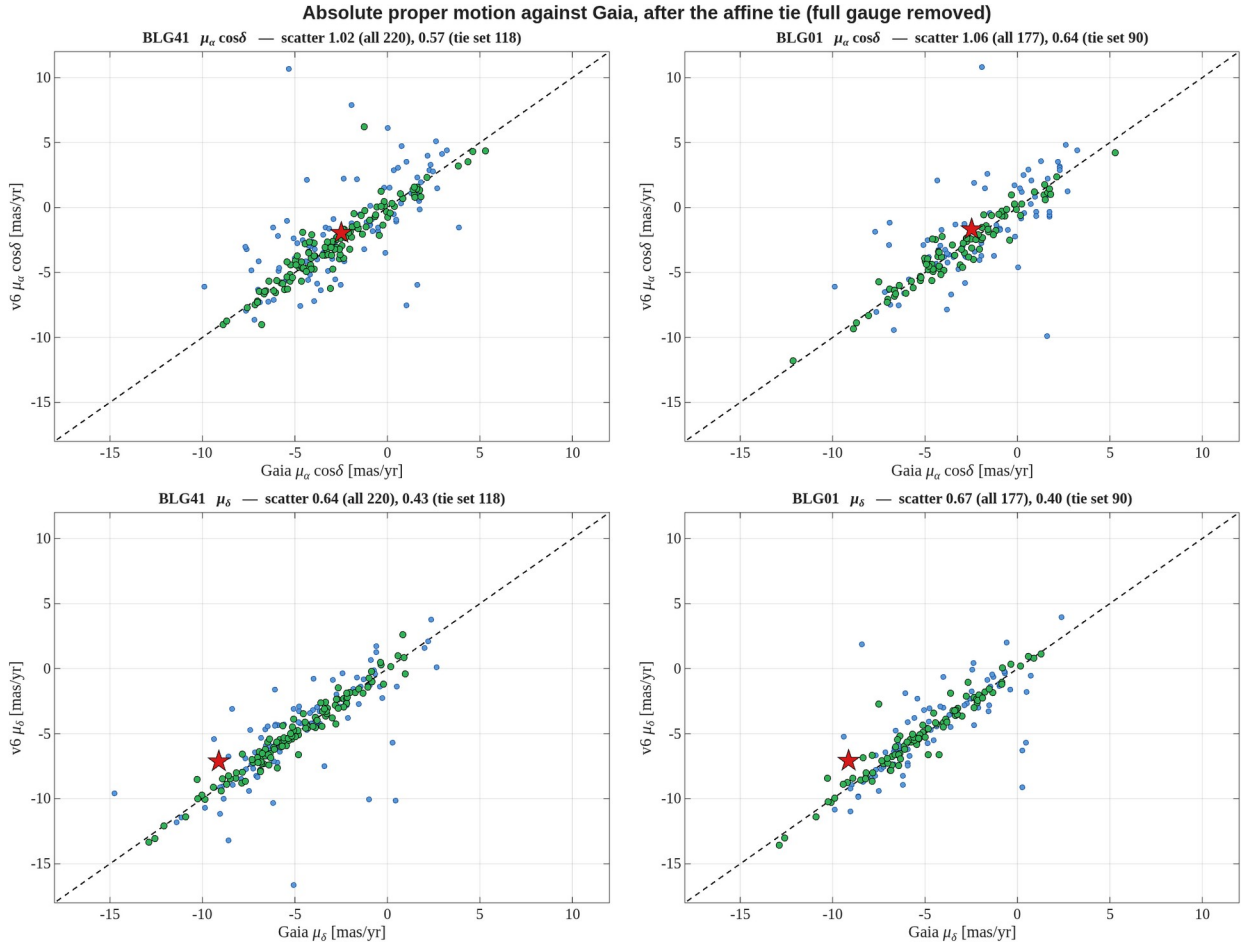


Figure 2. Our absolute proper motion against Gaia's, per axis and per field, after the affine tie with the **full six-parameter gauge** removed. Dashed line is 1:1. The scatter quoted in each panel is the robust standard deviation of the difference; it improves by a factor of three to four over a tie that removes only the constant. The target is the red star.
report_v6_pm_vs_gaia.png

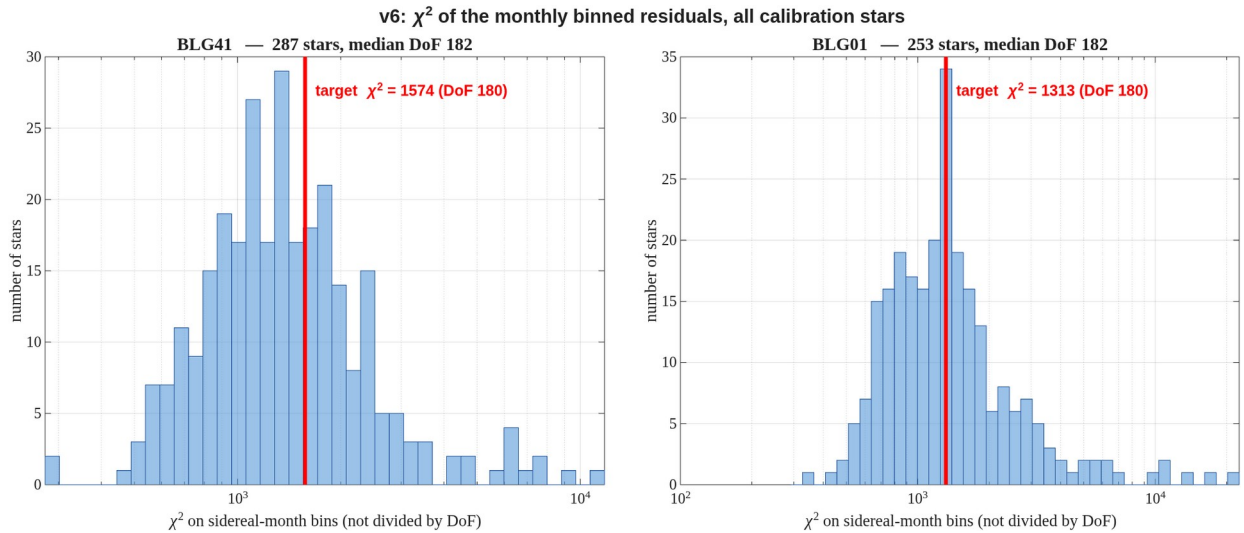


Figure 3. χ^2 of the sidereal-month binned residuals for every calibration star, both axes combined, **not divided by DoF**. Each bin contributes (mean / standard error)² per axis and DoF = $2N_{\text{bins}} - 4$. The red line marks the target, which sits in the body of the distribution rather than the tail.
report_v6_chi2.png

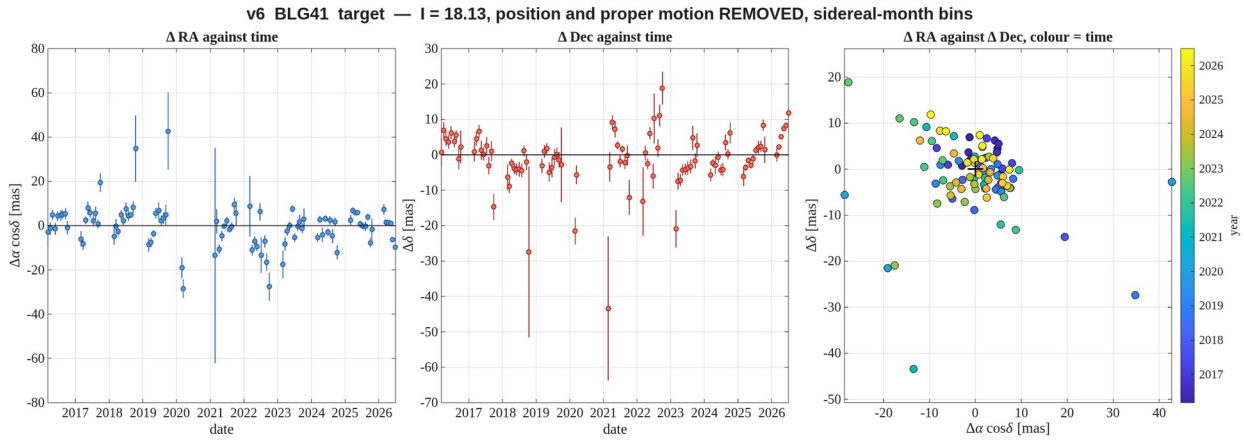


Figure 4. **BLG41** — the target, $l = 18.13$. **Position and proper motion removed:** these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
report_v6_radec_BLG41_target.png

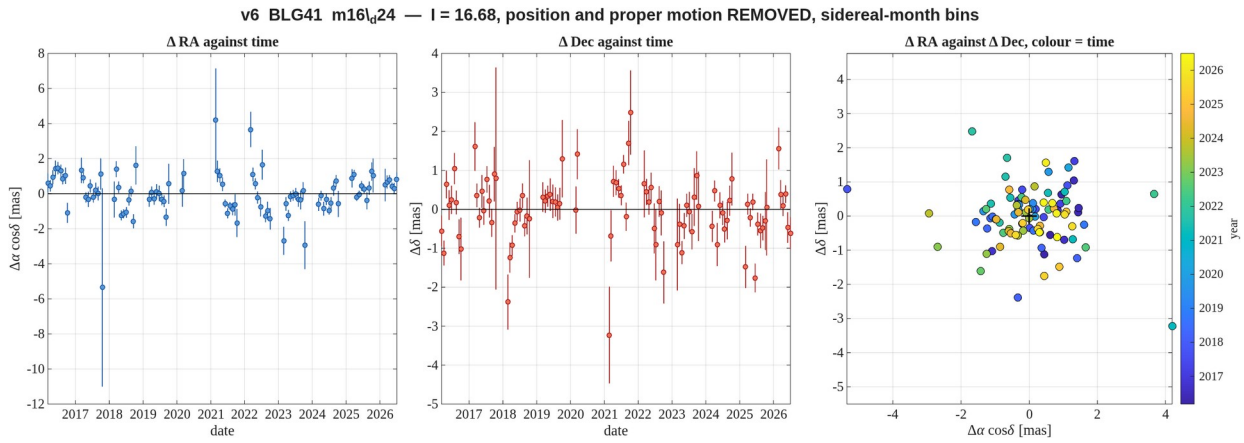


Figure 5. **BLG41** — comparison star $l = 16.68$, 24 pix from the target — the best measured of the set. **Position and proper motion removed:** these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
report_v6_radec_BLG41_m16_d24.png

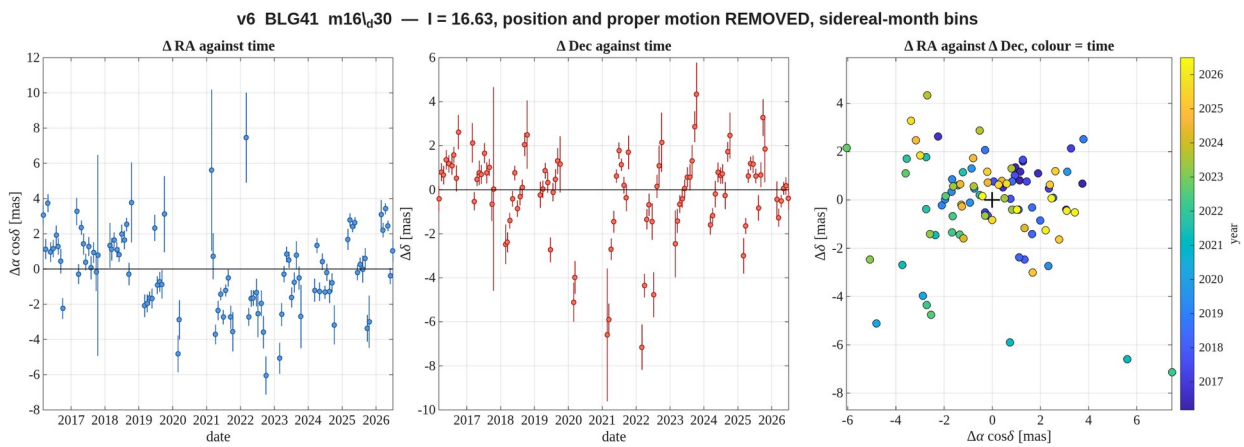


Figure 6. **BLG41** — comparison star $l = 16.63$, 30 pix from the target. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
report_v6_radecl_BLG41_m16_d30.png

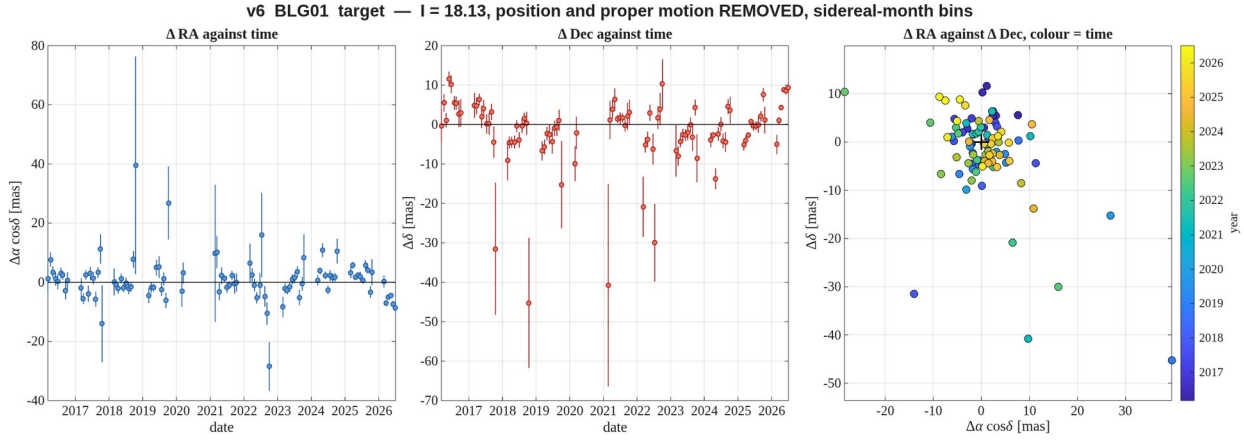


Figure 7. **BLG01** — the target, $l = 18.13$. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
report_v6_radecl_BLG01_target.png

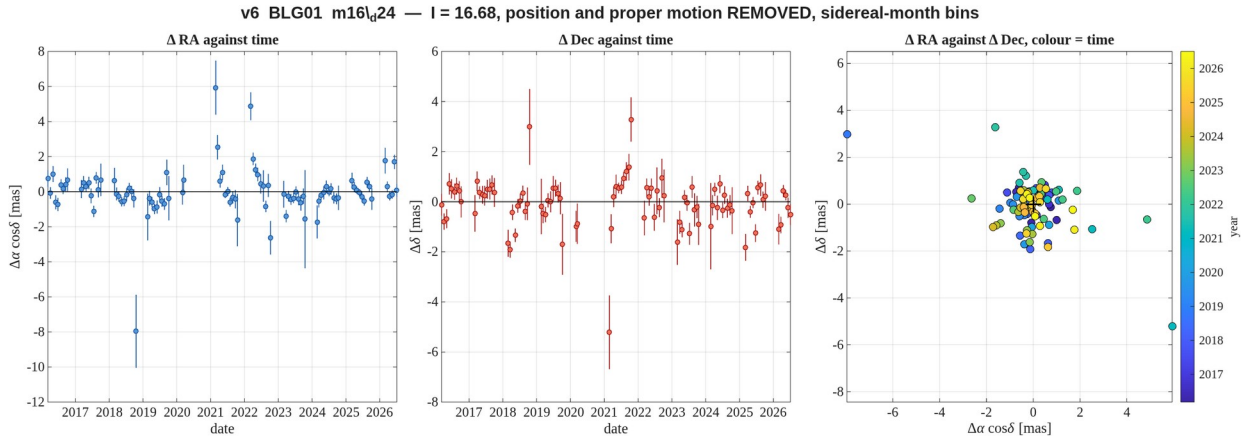


Figure 8. **BLG01** — comparison star $l = 16.68$, 24 pix from the target — the best measured of the set. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
report_v6_radecl_BLG01_m16_d24.png

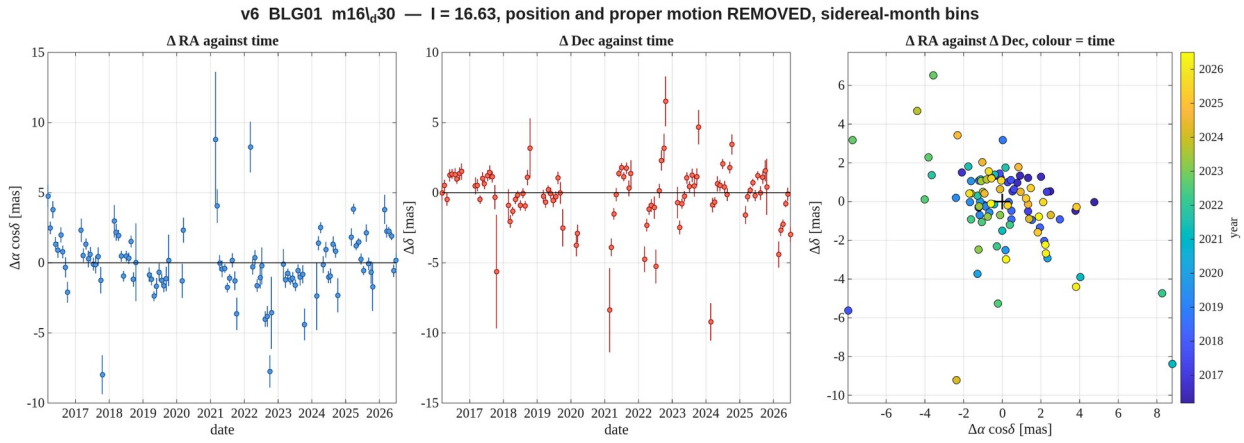


Figure 9. **BLG01** — comparison star $l = 16.63$, 30 pix from the target. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.

report_v6_radec_BLG01_m16_d30.png

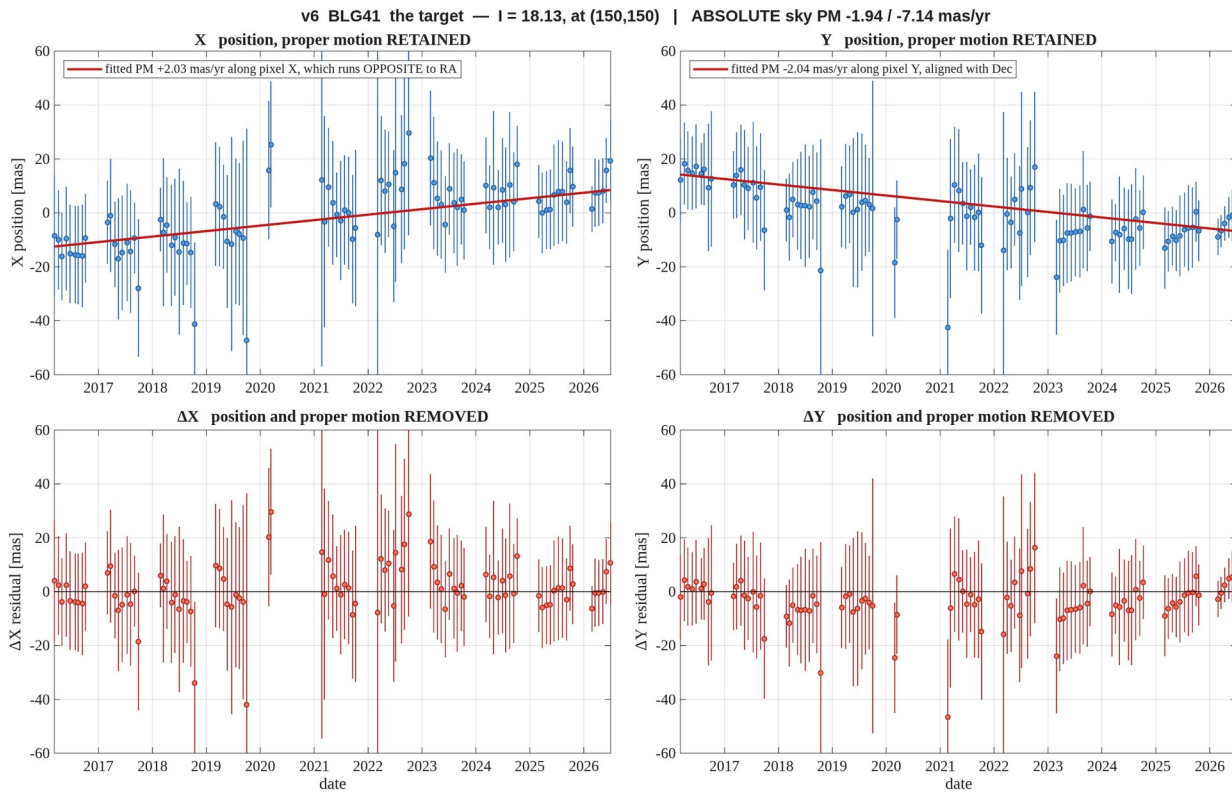


Figure 10. **BLG41** — the target, $l = 18.13$. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_target.png

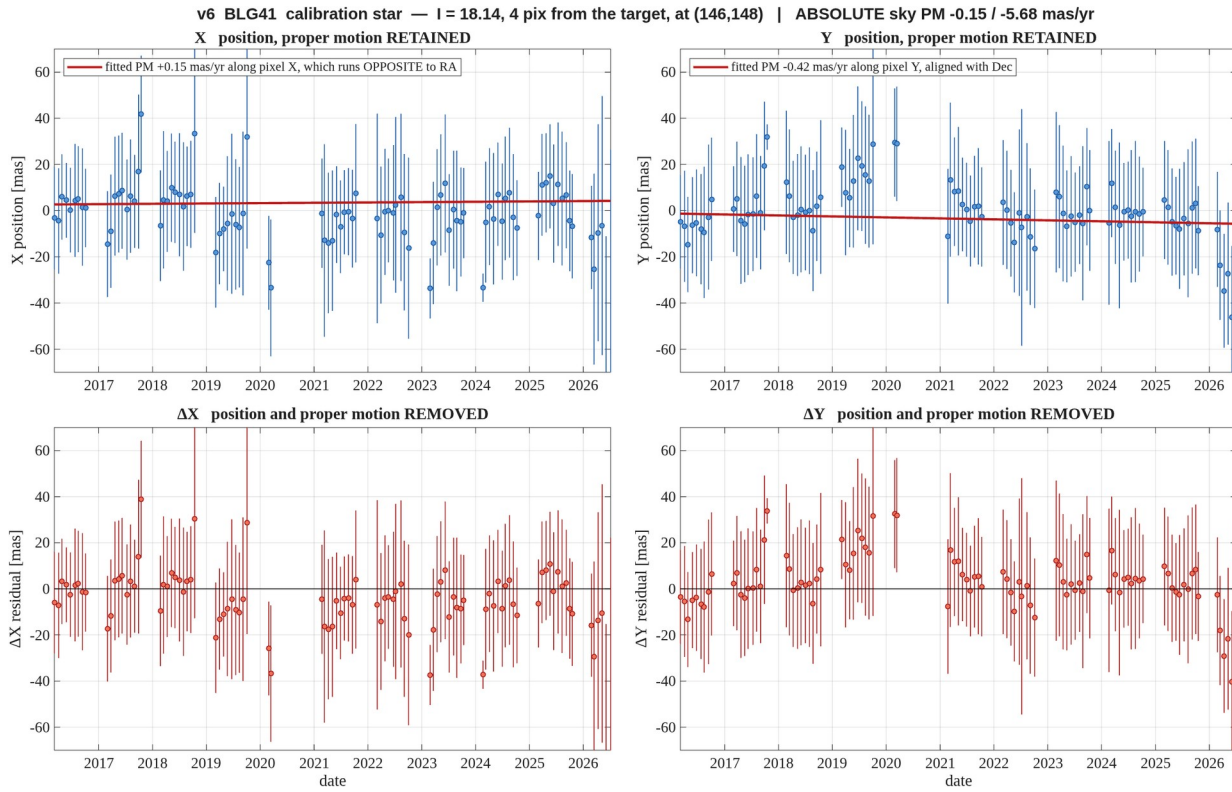


Figure 11. **BLG41** — $l = 18.14$, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_m18_d04.png

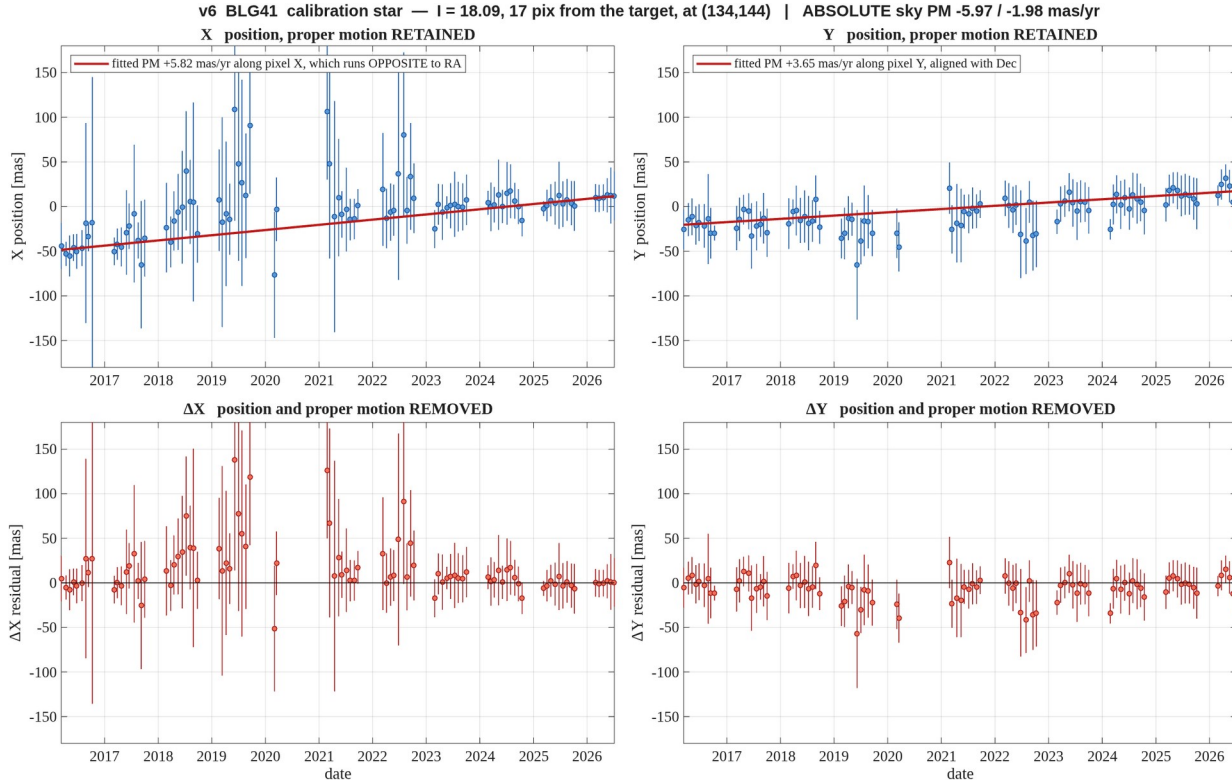


Figure 12. **BLG41** — $l = 18.09$, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_m18_d17.png

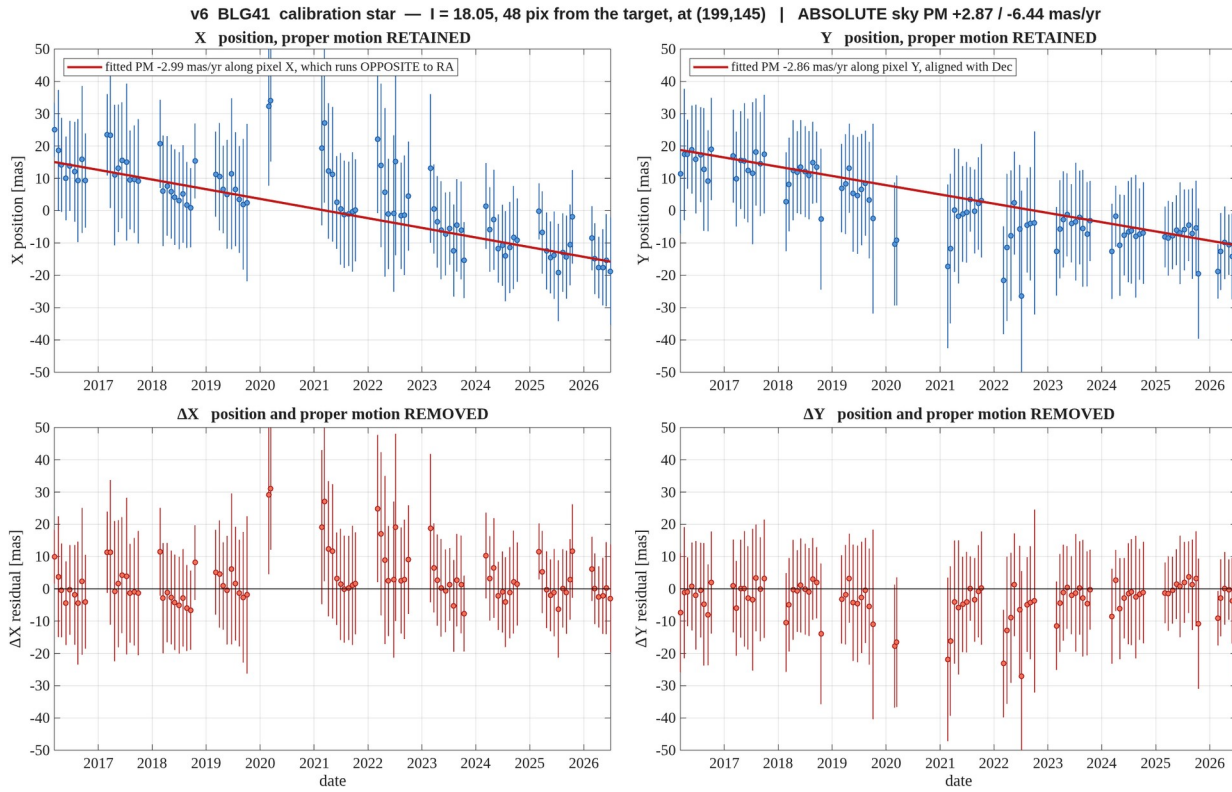


Figure 13. **BLG41** — $l = 18.05$, 48 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_m18_d48.png

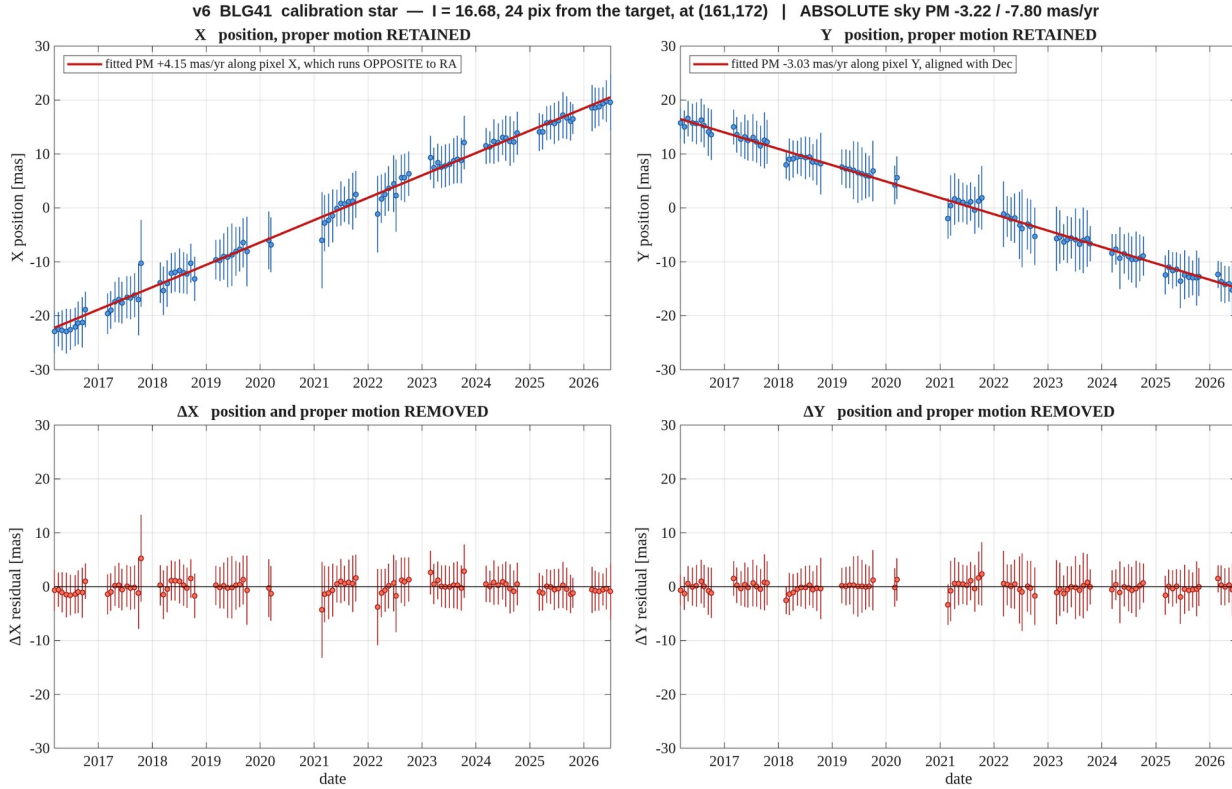


Figure 14. **BLG41** — $l = 16.68$, 24 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_m16_d24.png

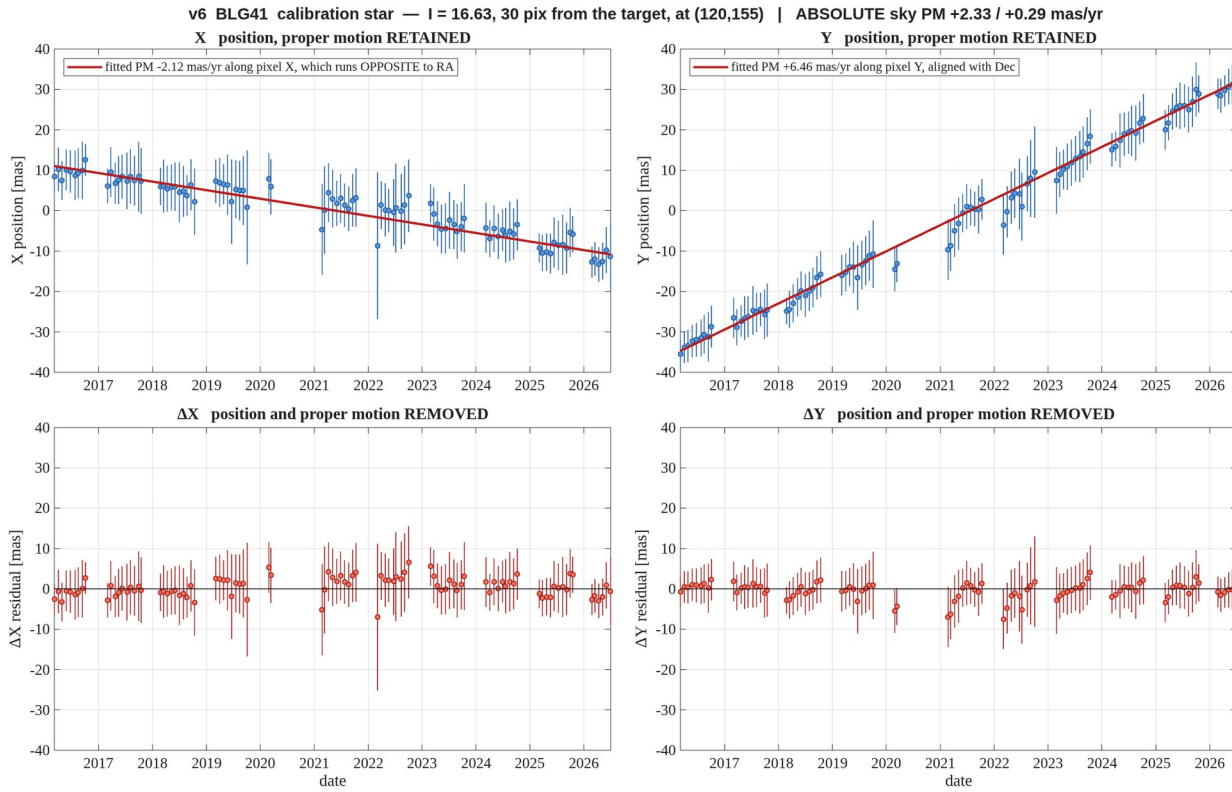


Figure 15. **BLG41** — $l = 16.63$, 30 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_m16_d30.png

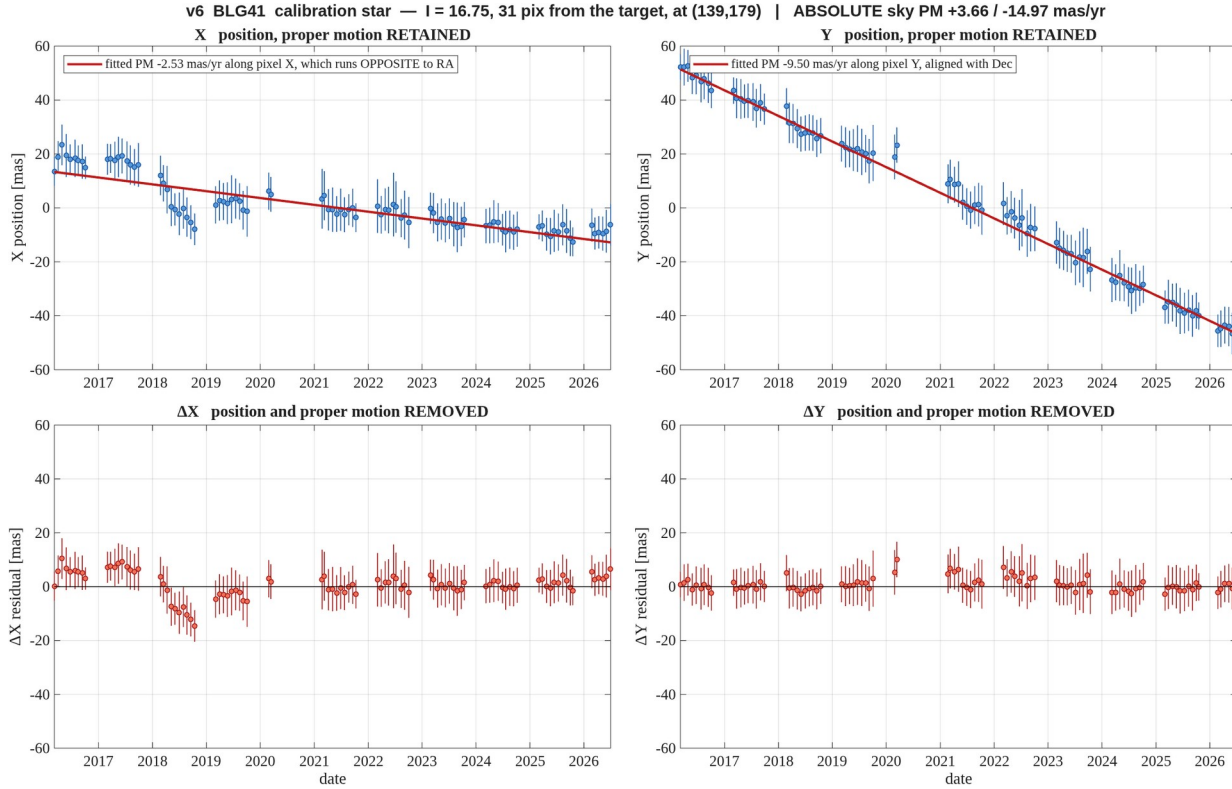


Figure 16. **BLG41** — $l = 16.75$, 31 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG41_m16_d31.png

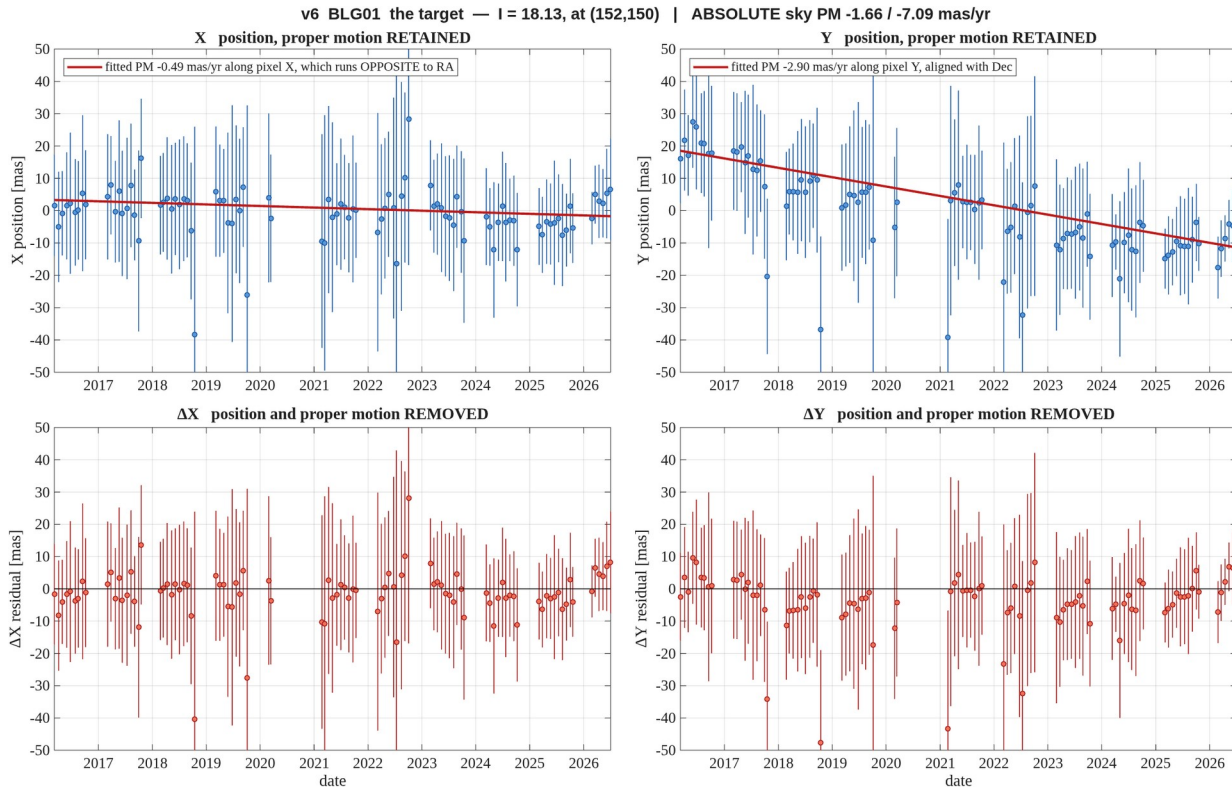


Figure 17. **BLG01** — the target, $l = 18.13$. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG01_target.png

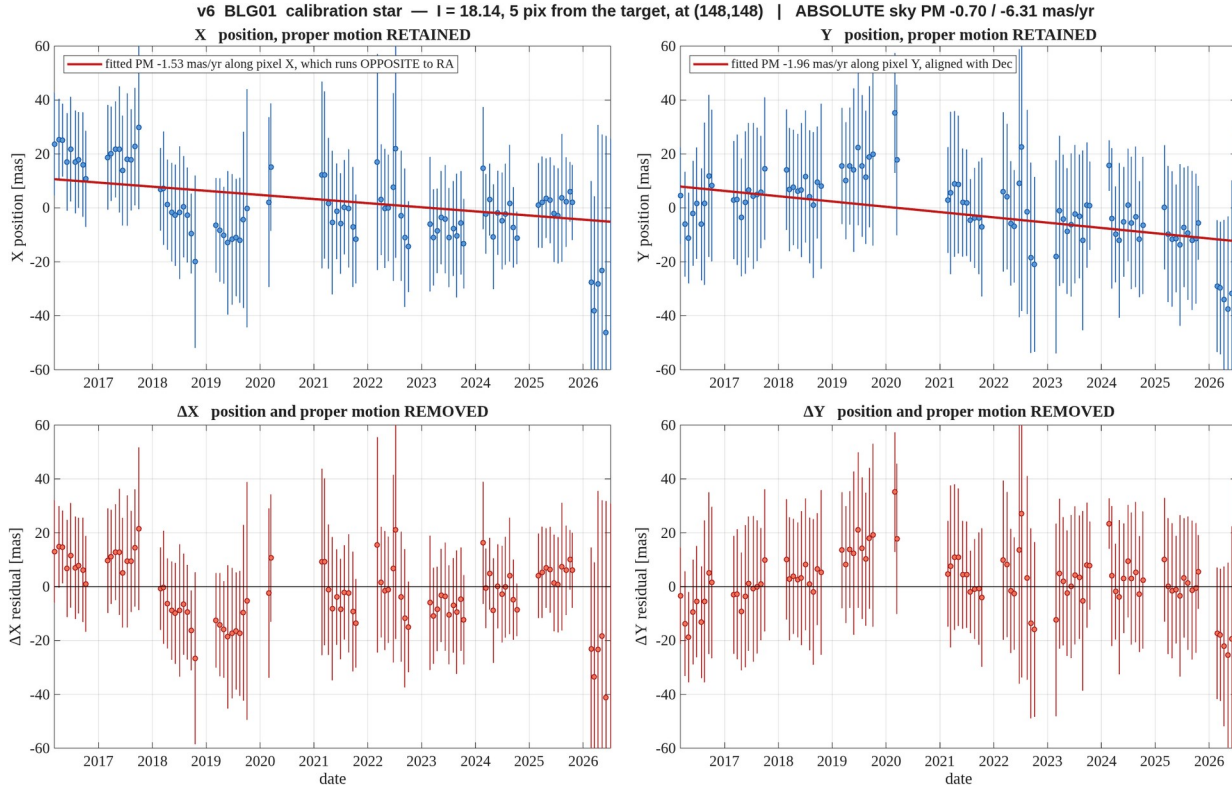


Figure 18. **BLG01** — $l = 18.14$, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG01_m18_d04.png

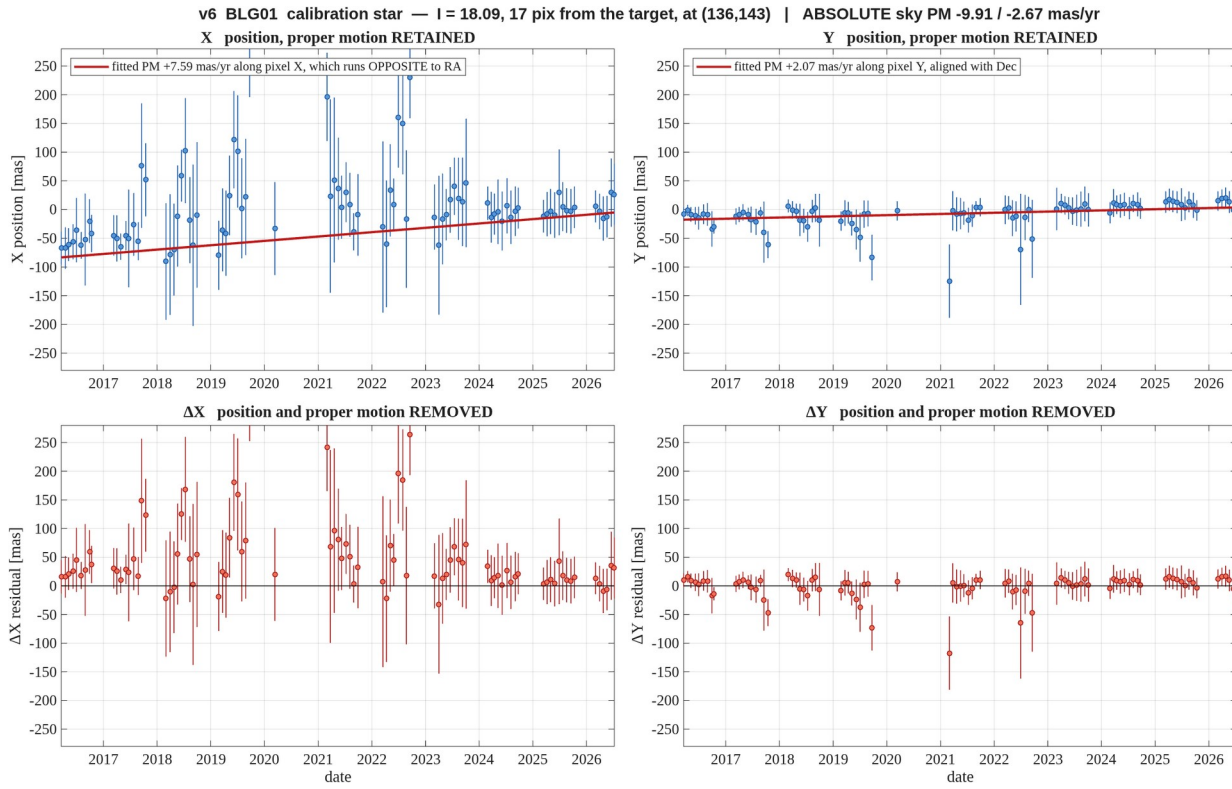


Figure 19. **BLG01** — $l = 18.09$, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG01_m18_d17.png

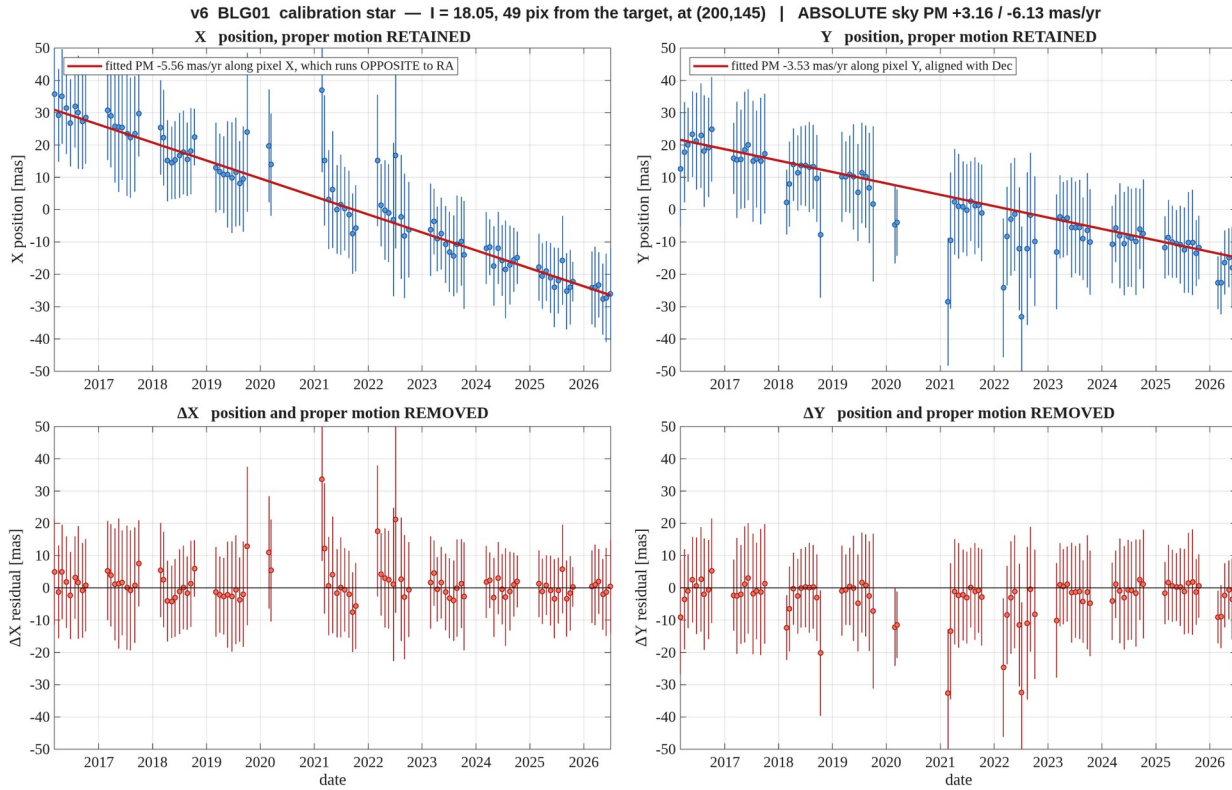


Figure 20. **BLG01** — $l = 18.05$, 48 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG01_m18_d48.png

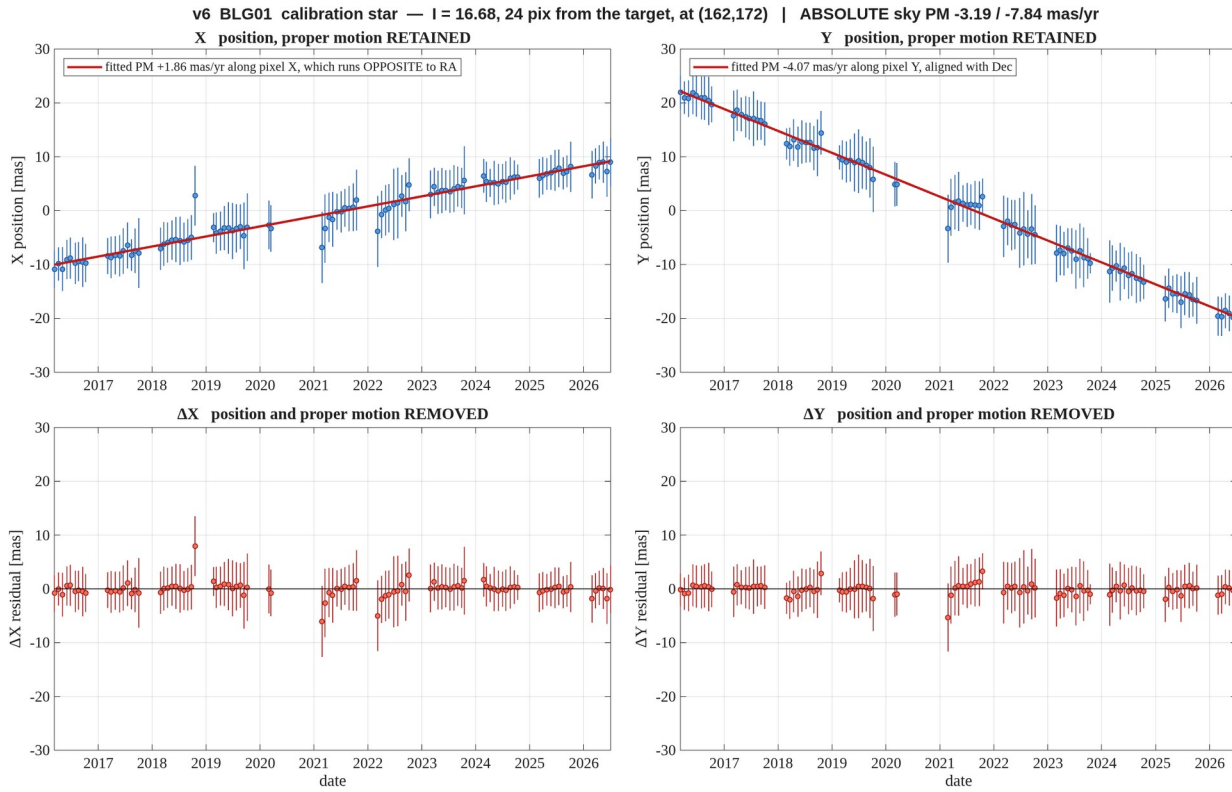


Figure 21. **BLG01** — $l = 16.68$, 24 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG01_m16_d24.png

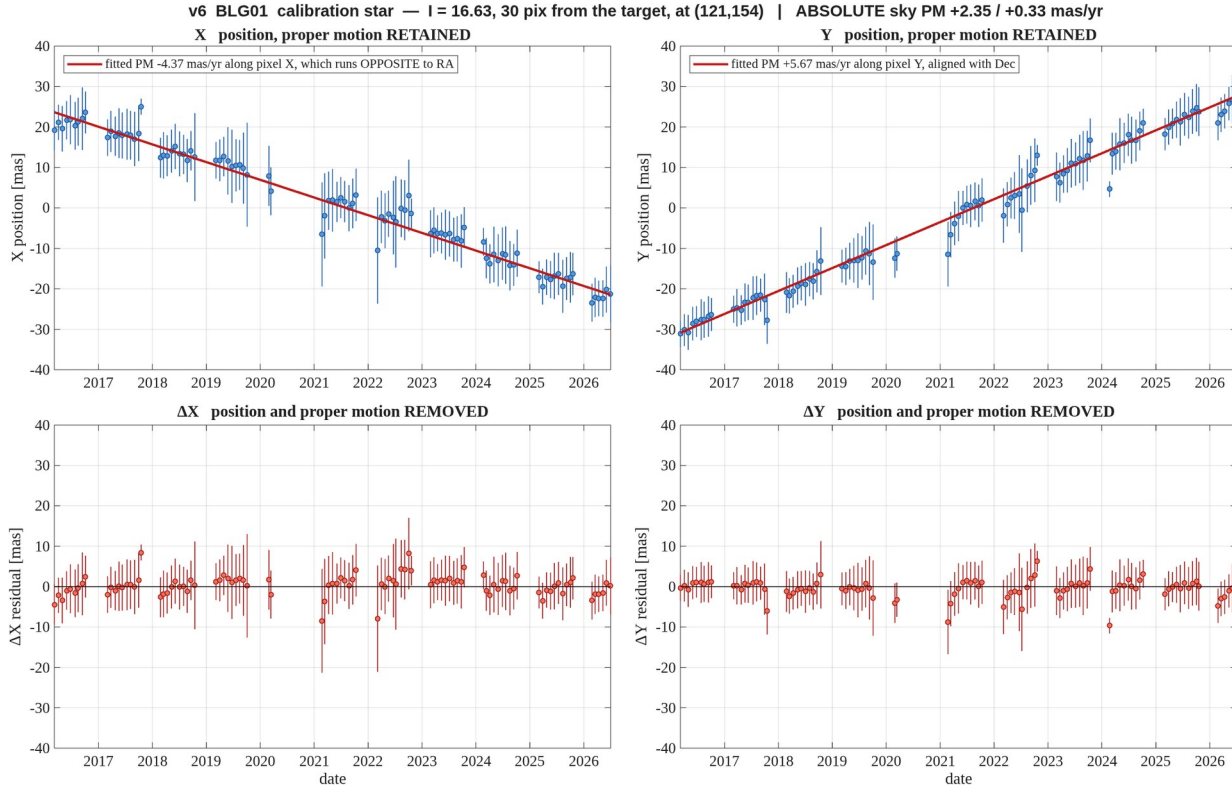


Figure 22. **BLG01** — $l = 16.63$, 30 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v6_motion_BLG01_m16_d30.png

